STATIC NAT CONFIGURATION & VERIFICATION

Cisco Packet Tracer Technical Implementation & Validation Report

Topology:	3 PCs, Switch (SW1), Router (R1), Cloud Internet	**Date:**	July 26, 2026
Target Device:	Cisco Router R1 (IOS)	**Status:**	PASSED / VERIFIED

1. Executive Overview

This technical document details the implementation and testing of **Static Network Address Translation (Static NAT)** on Cisco Router `R1`. The primary goal of Static NAT in this network architecture is to establish a permanent one-to-one mapping between internal private IPv4 addresses (`172.16.0.0/24`) and dedicated public IPv4 addresses (`100.0.0.x/24`), enabling bidirectionally routable Internet access.

2. Network Topology & Addressing Schema

The network comprises an internal LAN segment connected to `G0/1` of `R1` and an external WAN/Internet connection on `G0/0` of `R1` leading to a remote DNS/Internet destination (`8.8.8.8`).

Device Name	Interface	Inside Local IP (Private)	Inside Global IP (Public Static)	NAT Role
PC1	NIC	172.16.0.1 /24	100.0.0.1 /24	Static Inside Source
PC2	NIC	172.16.0.2 /24	100.0.0.2 /24	Static Inside Source
PC3	NIC	172.16.0.3 /24	100.0.0.3 /24	Static Inside Source
R1 (Inside)	GigabitEthernet0/1	172.16.0.254 /24	N/A	`ip nat inside`
R1 (Outside)	GigabitEthernet0/0	203.0.113.1 /30	N/A	`ip nat outside`
Internet Host	Server	8.8.8.8	8.8.8.8	Remote Target

3. Lab Objectives

1. Configure the appropriate `ip nat inside` and `ip nat outside` interfaces on Router `R1`.
2. Create static 1-to-1 NAT mappings for `PC1`, `PC2`, and `PC3` to public IPs in the `100.0.0.x/24` subnet.
3. Verify end-to-end ICMP reachability from internal hosts to `8.8.8.8`.
4. Inspect and analyze active NAT translation tables and statistics on `R1`.
5. Test domain name resolution (`google.com`) across translated sessions.

4. Router R1 Configuration Commands

The following commands were executed in global configuration mode on Router `R1` to assign interface roles and configure static translation entries:

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#int g0/0
R1(config-if)#ip nat outside
R1(config-if)#ex
R1(config)#int g0/1
R1(config-if)#ip nat inside
R1(config-if)#ex
R1(config)#ip nat inside source static 172.16.0.1 100.0.0.1
R1(config)#ip nat inside source static 172.16.0.2 100.0.0.2
R1(config)#ip nat inside source static 172.16.0.3 100.0.0.3
R1(config)#ex
%SYS-5-CONFIG_I: Configured from console by console
```

5. Reachability Verification & Testing Results

Test 1: PC1 Ping Validation

Initial connectivity test from PC1 to Internet host `8.8.8.8` experienced normal initial packet loss due to ARP table resolution across the gateway path, followed by 100% success on the subsequent attempt.

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=52ms TTL=126
Reply from 8.8.8.8: bytes=32 time=12ms TTL=126
Reply from 8.8.8.8: bytes=32 time=12ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 52ms, Average = 19ms
```

Test 2: PC2 & PC3 Ping Validation

Subsequent pings from PC2 and PC3 to `8.8.8.8` executed with zero packet loss, confirming active static translation and immediate forwarding:

```
PC2 Command Output:
Reply from 8.8.8.8: bytes=32 time=10ms TTL=126
Reply from 8.8.8.8: bytes=32 time=14ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Ping statistics for 8.8.8.8: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

```
PC3 Command Output:
Reply from 8.8.8.8: bytes=32 time=15ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Ping statistics for 8.8.8.8: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

6. Router NAT Table & Statistics Inspection

1. Static NAT Mappings & Initial Statistics

Executing `show ip nat translations` and `show ip nat statistics` on `R1` confirms the 3 configured static bindings and verifies interface assignments:

```
R1#sh ip nat trans
Pro  Inside global     Inside local       Outside local      Outside global
---  100.0.0.1         172.16.0.1         ---                ---
---  100.0.0.2         172.16.0.2         ---                ---
---  100.0.0.3         172.16.0.3         ---                ---

R1#sh ip nat statistic
Total translations: 3 (3 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/0
Inside Interfaces: GigabitEthernet0/1
Hits: 4  Misses: 4
Expired translations: 4
Dynamic mappings:
```

2. Dynamic Translation Entries During Live Traffic & Web Query

When active traffic passes through `R1` (ICMP pings and UDP DNS lookup for `google.com`), `R1` dynamically binds protocol ports while preserving the configured **Inside Global** IP addresses:

```
R1#sh ip nat translation
Pro  Inside global     Inside local       Outside local        Outside global
icmp 100.0.0.2:1       172.16.0.2:1       8.8.8.8:1            8.8.8.8:1
icmp 100.0.0.2:2       172.16.0.2:2       8.8.8.8:2            8.8.8.8:2
icmp 100.0.0.2:3       172.16.0.2:3       8.8.8.8:3            8.8.8.8:3
icmp 100.0.0.2:4       172.16.0.2:4       8.8.8.8:4            8.8.8.8:4
icmp 100.0.0.3:1       172.16.0.3:1       8.8.8.8:1            8.8.8.8:1
icmp 100.0.0.3:2       172.16.0.3:2       8.8.8.8:2            8.8.8.8:2
icmp 100.0.0.3:3       172.16.0.3:3       8.8.8.8:3            8.8.8.8:3
icmp 100.0.0.3:4       172.16.0.3:4       8.8.8.8:4            8.8.8.8:4
---  100.0.0.1         172.16.0.1         ---                  ---
---  100.0.0.2         172.16.0.2         ---                  ---
---  100.0.0.3         172.16.0.3         ---                  ---

R1#sh ip nat translation
icmp 100.0.0.1:10      172.16.0.1:10      172.217.175.238:10   172.217.175.238:10
icmp 100.0.0.1:11      172.16.0.1:11      172.217.175.238:11   172.217.175.238:11
icmp 100.0.0.1:12      172.16.0.1:12      172.217.175.238:12   172.217.175.238:12
icmp 100.0.0.1:9       172.16.0.1:9       172.217.175.238:9    172.217.175.238:9
---  100.0.0.1         172.16.0.1         ---                  ---
---  100.0.0.2         172.16.0.2         ---                  ---
---  100.0.0.3         172.16.0.3         ---                  ---
udp  100.0.0.1:1025     172.16.0.1:1025    8.8.8.8:53           8.8.8.8:53
```

Technical Observation: DNS & Web Query Translation

Notice in the second translation table output above that `PC1` (`172.16.0.1`) sent a DNS query on UDP port `1025` to DNS server `8.8.8.8:53`. The router translated the source IP to `100.0.0.1`. Subsequent ICMP packets directed to `172.217.175.238` (resolved IP for `google.com`) were also successfully translated and routed.

7. Summary & Key Takeaways

- **Permanent 1:1 Mapping:** Static NAT permanent mappings remain active in the translation table regardless of whether traffic is currently flowing.
- **Bidirectional Translation:** Because the mappings are static, outside hosts can initiate connections to internal hosts using their Inside Global IP (e.g., `100.0.0.1`).
- **Interface Designation:** Proper configuration of `ip nat inside` on LAN interfaces and `ip nat outside` on WAN interfaces is mandatory for Cisco IOS to identify when translation is required.